A Brief Introduction to Protein-ligand Docking

Basic Principles and Applications

(October 2020)

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Preface

Expectations? Reality?

Your expectations?

1. What kind of “research experience” you are looking for?
2. What you expect a computer can do?

Reality

1. What kind of “research experience” you are looking for?
   - Research =/= achievement
   - Academia =/= let you hide from the job market
2. What you expect a computer can do?
   - Main advantages:
     - Virtually screen and design inhibitors → reduce experimental cost
     - Study phenomena that are difficult to be investigate experimentally
   - You think its easier to handle than experimental work?
     - NO! the learning curve can be much steeper
   - **LIMITS: Computer cannot do everything**

     →**Still have false positive/false negative hits!!!!!**

     →**Note the assumptions you are making!!!!!!**

     →**Always try to validate your results if possible!!!!!**

     →**Be VERY careful with the predicted/generated results!!!!!!**

Preparation

What you need to do before you perform docking

Importance of literature review

- Function of enzyme
- Catalytic mechanism
- pH of environment: physiological pH? Acidic (e.g. stomach)?
- Protonation especially around the binding site
- metal coordination
- ions
- water
- Membrane protein?

Identify the purpose of docking

- MD input preparation of unknown molecule in known receptor
- virtual screening of novel inhibitors
- brief insights of Inhibition mechanism
- QSAR
- …

Selection of receptor starting structure

- apo vs ligand bonded
- Complexed metal ions
- mutations?
- resolution

Selection of binding site to be targeted

- Approaches
  - choosing a known pocket
  - pocket detection
  - blind docking
  - notes on allosteric site: choice of structures

Selection of the ligands to be modelled

- Complexed ligand
- Active compounds from literature without a crystal structure
- Ligands from a similarity/pharmacophore search
- Virtual screening of a molecule database
- Ligands that you design yourself

Basic concepts of docking

A very brief introduction to the theories behind molecular docking

General

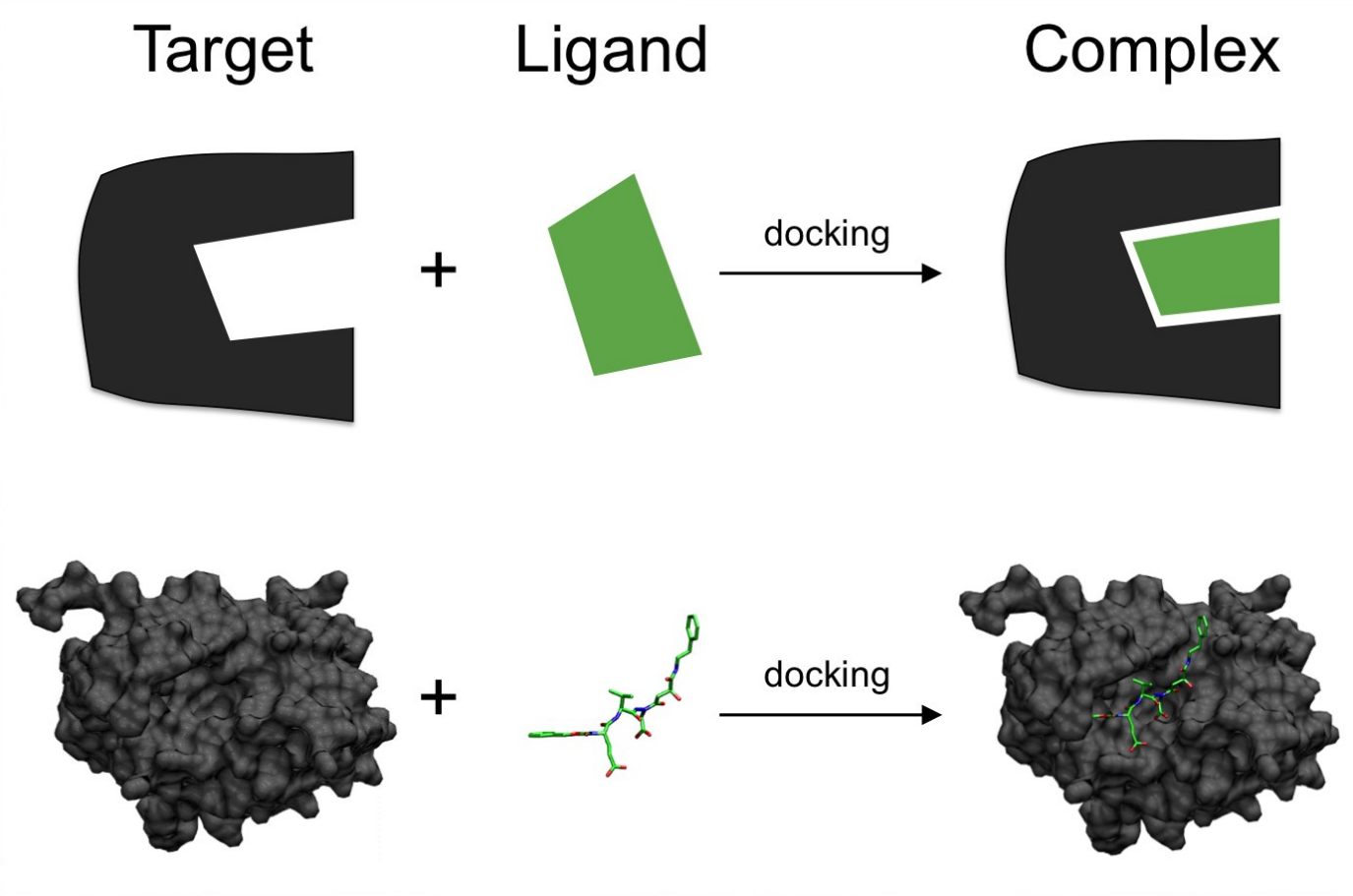


Image adapted from Wikimedia commons

Conformational search

- Conformational search v.s. geometry optimization
- Methods of conformational search
  - Systematic vs random search
  - Simulation based method
  - Genetic algorithm

Scoring methods

- Examples
  - Force-field based
  - Alignment based/pharmacophore based
  - Combination of methods
- In many cases the scores are relative to system and many have limited transferability

Rigid docking, flexible docking and MD

- Speed vs accuracy
  - Rigid docking is still a useful tool after 20 years
- Assumptions:
  - with force-field based method assuming no reactions

Docking, redocking and cross docking

- Re-docking refers to the ability to reproduce co-crystallized binding geometry and orientation of the associated ligand given a rigid macromolecule state.
- Cross-docking refers to utilizing different ligand structures isolated from multiple PDB files of the same protein to test against a single rigid protein model structure.

Practical example

Software to be used

- Download links of AutoDock4 and VINA:
  - http://autodock.scripps.edu/downloads
- Download links of AutoDockTools:
  - http://mgltools.scripps.edu/downloads
  - Note: Windows user need to set the path to the startup directory
- Download links of UCSF Chimera:
  - https://www.cgl.ucsf.edu/chimera/download.html
- Small molecule builder: pymol, Avogadro (avogadro.cc)

Receptor preparation

- Pre-processing with UCSF Chimera
- Separate ligand from receptor
- Remove unwanted nonbonded residues
- Adding missing residues
- Adding missing atoms
- Protonation: fast vs more accurate
  - Servers for predicting protein protonation at specific pH
    - H++ server (need to register, sometimes can't load) http://biophysics.cs.vt.edu/
    - pdb2pqr server http://server.poissonboltzmann.org/pdb2pqr
- Using ADT GUI to generate the receptor .pdbqt
- Manually editing pdbqt
  - to add the charge of metal, only required for AD4.
  - Vina does not consider charges in the scoring function.

ADT GUI of AD4

- Setting up the grid box for AD4:
  - run autogrid
- Set AD4 parameters
- Run AD4
- Visualization of AD4 results in ADT and interpretation

ADT GUI of VINA

- Setting up the grid box
- Set VINA parameters
- Running VINA via GUI
- Visualization of VINA results in ADT

Other GUIs of VINA

- UCSF Chimera interface (limited functionality)
  - Directly visualize VINA results in UCSF Chimera:
    - instant and higher quality graphics for Windows, loads faster, although less interaction to be visualize automatically (need extra steps)
  - Interface to a local copy of VINA
    - Limits:
      - Cannot set the number of rotatable bonds of ligands
      - Cannot set flexible residues for receptors
      - Limited choice of docking parameters
      - Limited ability of assigning charges to non-standard residues of receptor
    - Advantages:
      - More convenient
      - Macromolecules loads faster than ADT
      - Easier to compare the docking pose in different receptors
      - Alternative GUI of VINA if you use the latest MacOS that ADT does not support

The command line interface

- Utility scripts for doing everything without a GUI
- Sample usage

Virtual screening with ADT cmd tools

- Automatic input -> docking with minimal scripting
- Possible to pipe to MD

Reference

- Leach, A. *Molecular Modelling*; Pearson Education (US), 2001-01-15.
- AutoDock4 + ADT tutorial
  - http://autodock.scripps.edu/faqs-help/tutorial/using-autodock-4-with-autodocktools
- More updated & practical ADT tutorials from Bevan & Brown's Lab
  - https://osf.io/82n73/wiki/Docking%2C%20Re-docking%2C%20and%20Cross%20Docking/